

I: Infinite Go

This chapter it's a draft, at least for now. I'll include my thought process and most of what I'm able to understand/formalize about it.

To begin with, let's see *where* we are playing now. Our infinite board should be able to have also infinite dimensions and every one of those could or not be infinite as well.

We'll begin with the simplest case (I hope), let's play on a two dimensional board that has one infinite dimension.

Let Γ be a Go game such that,

$$G = \{1, 2, \dots, n\} \times \mathbb{Z}.$$

for $n \in \mathbb{N}$. We'll use $\mathbb{Z}$ for simplicity as it's a better (easier) representation of the infinite board. As the number of players does not matter at all, let's use only two players and, the canonical adjacency. Our game will be $\Gamma = (G, E, A_{ort}, \Delta_{Go_{\infty}})$. At least for now, we are not sure how $\Delta_{Go_{\infty}}$ works but, we will get there.

Our board could (and for us, will) look like this, Figure 1.1. Now, there is a substantial change in this new game.

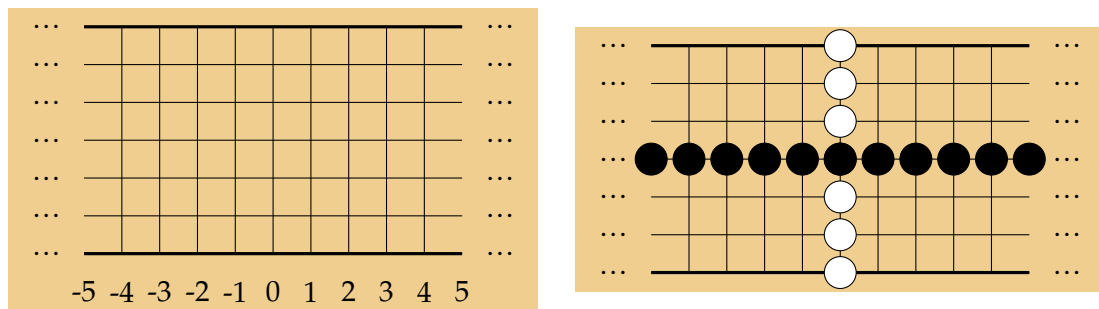


Figure 1.1 – A $7 \times \infty$ board and two intersecting plays that resolves by definition.

Notice that, if the white line were infinite, such play generated two infinite lines in one turn, which is crazy.

The very first problem on the infinite board it's how to determine a capture. Any other prior definition should work just fine.

Lets say we are in the next position, Figure 1.2

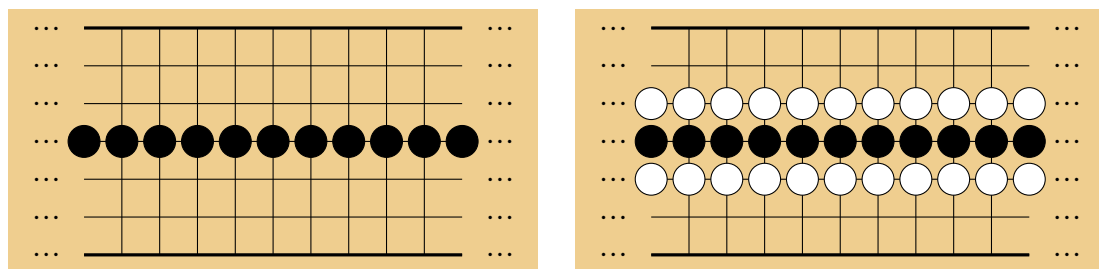


Figure 1.2 – An infinite board with an infinite black group being capture.

As this group is infinite (numerable), will touch the very end of the board so we can capture it by surrounding it at the top and bottom.

We can be sure that this is capture because if the black group it's I , then $K(I) \subset S$ where S is the set of the two white groups. That is, $L(I)$ should be 0. Now let's take a look at another example, an infinite group but only on one side.

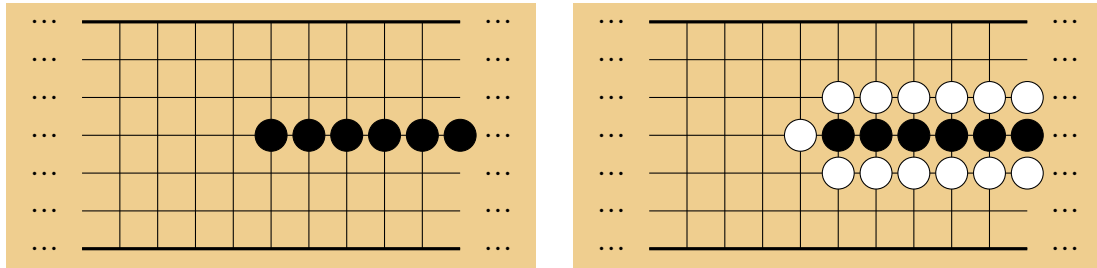


Figure 1.3 – An infinite, on one side, black group being capture.

In this case, not using the whole board, even though it's infinite, its better. As it provides one more liberty (or does not loses it).

The following example reflects and interesting nature of inifinite Go. (More on this later)

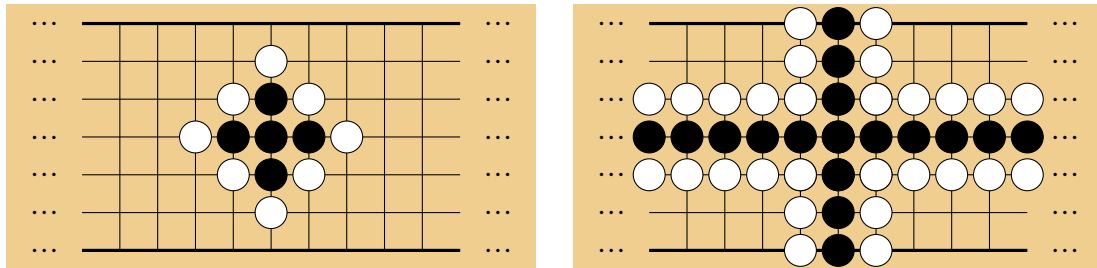


Figure 1.4 – Two captures.

It seems that finite groups are stronger than infinite ones as, again, they don't lose liberties due to not being infinite.

Lets characterize *what* can be done in the game.

Definition 1.0.1 (A move). In an infinite game of Go of dimension n , a player in their turn t can perform one of the following actions:

1. Pass.
2. Propose a connected hyperplane H of dimension at most $n - 1$. The hyperplane H is a subset of the board such that, for at least one dimension, all its points have a fixed coordinate. All the stones in H have the color of the player making the move. In particular, if $\dim H = 0$, then H is asingle stone.

The stones already present on the board may intersect H . Let

$$P = \{P(x) \in T_t : C(x) \neq 0\}$$

be the set of occupied points at turn t . The stones that are actually played are then

$$H - P = \{h \in H : P(h) \notin P\}.$$

The set $H \setminus P$ may consist of several groups, as the stones already present on the board may disconnect the proposed hyperplane.

If any group contained in $H - P$ is in suicide condition, the proposed move is rejected. Otherwise, the stones in $H - P$ are placed on the board, and captures are resolved according to the usual rules.

Let me explain why $n - 1$ dimensions.

If we look back at the game of Go, stones are placed on a two dimensional board. In our formulation, this means that we place a connected 0 dimensional subset of the board, that is, a single stone. In other words, the dimension of the object being played is $n - 2$ relative to the dimension of the board.

If we kept this relation, we would obtain a natural generalization for higher dimensional boards, but we would be unable to define a move in one dimensional Go, since this would require

$$n - 2 = -1.$$

Since a negative-dimensional object is not part of our construction, we instead impose 0 as the minimum dimension of a playable hyperplane. Thus, in an n dimensional game, a player may propose a connected hyperplane H satisfying

$$0 \leq \dim H \leq n - 1.$$

1.1 Infinite SuperKo

We cannot use Zobrist method for the infinite Go as we will need infinite many bits. We need a better method, let's try to make one.

If we remember the nature of the SuperKo rule, it's says that we cannot replicate a prior board via a stone play. Again, we are using this version of SuperKo (Positional).

Because of the rule, every game has to be a line, that is, as we cannot go back into the history of plays (at least it's a pass but will not take in consideration those). From the empty board to the very end, we have a straight forward path. Not even that but, because of the rule, the whole game of Go its a *tree*. That is, starting from the empty board, there are 362 possible plays (361 stone placements and a pass) and we cannot go back. From those, every possible outcome, that is, any end of a match can be found by following a line of unrepeatd boards.

Let's define a way to measure how close are two given boards are.

First, lets notice that, in the very beginning, there are exactly one region and not much else in the board. That is, before the first turn, $|J| = 1$ (the big empty region).

Now, the very first turn will make so $|J| = 1$ if the black player plays a stone or a finite line. If the player chooses to play an infinite one, then $|J| = 2$. Lets pretend it just played an infinite group, with that, Zobrist method or any method that uses the stone itself go nuts and can't be use here but, $|J|$ it just a little bigger, finitely bigger. Now, this is not perfect as there are infinite many ways to play an infinite group so we end up with $|J| = 2$. Before going forward, lets see an example.

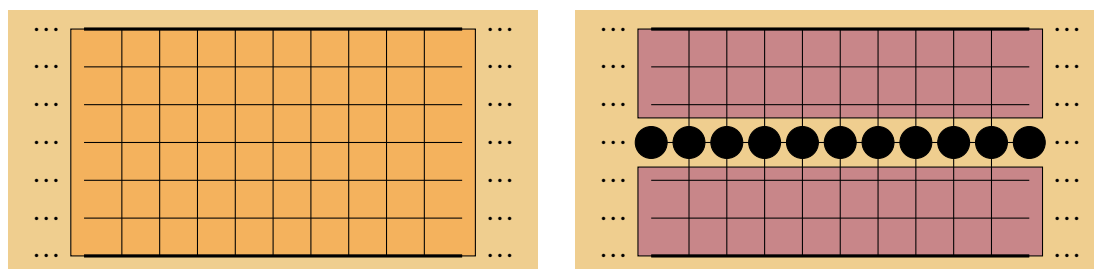


Figure 1.5 – A board where the number of regions (neutral and black) merely grows by one when an infinite group is played.

As I said early, playing such group vertically (in this case will be finite but you get the idea) or shifting it up or down, will end up with the same number of regions.

The first problem is to determine if the number of regions are always finite or not.

The second problem is how to determine those boards that has the same number of regions are infact different, we cannot look at them as that's too costly. My idea is, if we can solve the first problem, then we can measure every region and, by that, we can tell even if two boards have the same number of regions, those regions might not be equal (by biggest and smallest radius or something like that).

The third problem its color shifting, two boards might not be equal but the regions could be exactly the same (number and size), but the colors of the regions surrounding them might not. (I'm thinking about discrinating between finite and infinite regions. Then, in the set of finites, we can calculate the influence of each color, this should work but its just another filter and its far from perfect)

Lets attack the first problem.

In order to do it, we need to show something about the nature of the game, we'll characterize how to split a region and when that happens. For that, we need some tools first.

Definition 1.1.1 (Interior of a region). Given a region R , its interior R° are the points in R such that its adjacency set its in R , that is:

$$R^\circ = \{r \in R : K(r) \subset R\}$$

Lemma 1.1.1 (Adjacency bound). Given a $n \leq \infty$ dimensional board of Go, with the canonical adjacency, a point t in the board, has at most $2n$ adjacencies and at least n .

$$n \leq |t| \leq 2n.$$

Proof. This is left as an exercise for the reader. ●

Definition 1.1.2 (Dense hyperplane). Given a proposed hyperplane H , we say that H is dense if every stone $t \in H$ is occupied, that is,

$$C(t) \neq 0 \quad \forall t \in H.$$

Theorem 1.1.2. In an n -dimensional game of Go, the only moves that can split a region R of dimension d are dense hyperplanes of dimension $d - 1$.

Proof. ●

rough, but means that the only plays that we should care about are completely connected and "dense" that is they don't leave any empty spaces on the given plane, even that, only the ones that fill every single direction, except for the one we are unable to do so, are exactly the ones that splits a region.

Given an infinite game of Go, the number of regions are always finite? If there are infinite moves then the answer is no.

In order to see this, lets consider the infinite one dimensional Go.

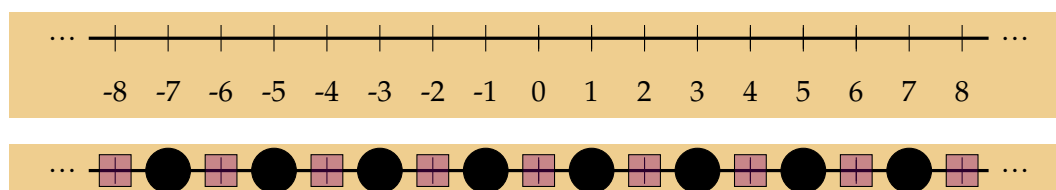


Figure 1.6 – A one dimensional infinite Go board and a example of infinite many regions by infinite many plays.

Is it possible to do many examples like this that end up with infinite many regions. The trick here and, perhaps, my salvation, is that there are infinite plays, which is, given the nature of the game, something that shouldn't happen. If we restrict ourselves into only finite many plays, then our question becomes:

Given an infinite game of Go, the number of regions generated by a finite number of plays are always finite too? More precisely, we want to prove that,

$$t < \infty \Rightarrow |J(T_t)| < \infty.$$

I'll give a shot via induction.

Theorem 1.1.3 (Finite regions induced by finite plays). *Given an infinite game of Go, the number of regions generated by a finite number of plays are always finite.*

Proof. ●

We are going to do an induction in the number of dimensions and the number of turns. if t its for turns and n its for dimensions, we'll have the following the game.

for $n = 1, t = 1$, we have at most two regions as we describe, we can only play single stones here. Then the placed stone splits the region into two.

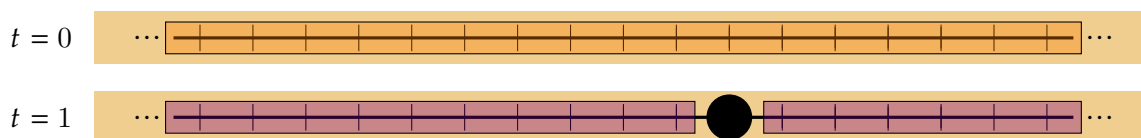


Figure 1.7 – An infinite one dimension and the first turn that shows how a region is split.

Lets assume that for the turn $t - 1$, the number of regions is finite.

for the t turn, the proposed play, given its legal, occupies at most every region. if p its the number of regions at turn t , then we at most can split every region, capture a finite number of groups (give we have play a finite number of turns). Therefore, the amount of regions stills finite.

now lets assume that for a $n - 1$ go board, the number of regions stills finite.

$n = n, t = 1$.

Again, the board begins empty and there is exactly one region, by playing whatever hyper-plane, the number of regions stills finite.

Lets assume that, for n dimensions and $t - 1$ turns, the number of regions stills finite.

For the t turn, we can only play in such way we touch at most a finite number of regions, the ones on turn $t - 1$. Lets suppose we can interact with every region. The result of such play will end up creating new regions, by capturing. Filling regions (assuming legality) or splitting them if we join an existent group. Finally the number of them stills finite.

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